A Depth-Averaged 2-D Physically-Based Model of Cohesive Dam/Levee Breach Processes

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ABSTRACT

A depth-averaged 2-D numerical model has been developed to simulate dam and levee breaching processes due to over-topping flows. The model solves the depth-averaged 2-D flow and sediment transport equations by using an explicit finite volume method. It considers interactions among transient flow, strong sediment transport and rapid bed change by including bed change and variable flow density in the flow continuity and momentum equations. The model adopts the HLLC approximate Riemann solver to handle the mixed-regime flows near the breach and the wetting-drying problem efficiently. The surface erosion rate method is used for cohesive sediment transport. The Widening mechanism is implemented by avalanche approach. Comparing with the USDA-ARS large-scale test, the model simulated value predict is better than former model.

Keywords: Dam and levee break and breaching, Cohesive sediment transport, Depth-averaged two-dimensional model, Finite-Volume method, lateral erosion.

INTRODUCTION

This now is brief of Introduction. Deambreak flow is significant problem for last decades because it will cause the the life and properties of people. Therefore, the development of Dambreak model is meaningful and inevitable. Nowadays, according to the dimensions, the Dambreak model is divided to 1D, 2D and 3D model. The 1D Dambreak model is popular in long-time, large scale case due to its scheme's simplicity and faster. However, 1D model is not cannot handle the complicated flow conditions such as the velocity and discharge in other than streamwise direction. In the contrast, 3D Dambreak model is time-consuming but can deal with these conditions since it consider three dimensions. consequently, 3D model is preferred in in short-time, small scale but complicate cases. 2D model's advangtages and disadvantages are between 1D and 3D model.

Recently, 2D model has been applied to simulates the flow overtopping dam and levee(Wu Weiming et al. 2012). However, in cohesive material cases, these models encounter several issues which need to be further investigated. First, in the cohesive material dam, the dam's slope is sharp, and cannot use shallow water equation which

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is valid under the assumption of gentle slope. Second, in the noncohesive material dam cases, the bed sliding happens when the bed's slope exceed the sediment's frictional angle. This is the mechanism of dam and levee getting wider in noncohesive dam. Nevertheless, it is case in cohesive material dam. Bed sliding does not happens when the bed's slope reach the frictional angle because of cohesive force. So the mechanism of widening of dam and levee is based on avalanche. Additionally, traditional model's vertical erosion can be problematic and should also consider the effect of lateral erosion.

The former depth-averaged 2D model's performance is not good enough in cohesive material dam because of the issues above. Hence, this paper consider the steep slope's effect in shallow water equation in the flow streamwise direction. Moreover, add the avalanche and laterial erosion mechanism to model. The governing equations, numerical methods and validation of this enhanced 2D-Dambreak flow model are illustrated in this paper. Last, this model is validated by the USDA-ARS large-scale test(Wu Weiming et al. 2012).

GOVERNING EQUATIONS

Shallow Water Equations

The shallow water equations are derived by depth-integration of Naiver-Stokes equations based on the hydrostatic pressure assumption which is derived from momentum equation in the vertical direction. The pressure assumption neglects the small velocity in the vertical direction that is valid in gentle slope bed but is problematic in steep slope. Therefore this paper introduced the effect of steep slope bed to the normal shallow water equations. The refined form of shallow water equation is derived as below

$$\frac{\partial(\rho h)}{\partial t} + \frac{\partial(\rho u h)}{\partial x} + \frac{\partial(\rho v h)}{\partial y} + \rho_b \frac{\partial z_b}{\partial t} = 0 \tag{1}$$

$$\frac{\partial(\rho v h)}{\partial t} + \frac{\partial(\rho u v h)}{\partial x} + \frac{\partial(\rho v^2 h)}{\partial y} + \cos^2\theta_x \rho g h \frac{\partial z_s}{\partial y} + \cos^2\theta_x \frac{g h^2}{2}\frac{\partial \rho}{\partial y} + \frac{\rho g n^2 U v}{\cos^{4/3}\theta_x h^{1/3}} = 0 \tag{2}$$

$$\frac{\partial(\rho v h)}{\partial t} + \frac{\partial(\rho u v h)}{\partial x} + \frac{\partial(\rho v^2 h)}{\partial y} + \cos^2\theta_x \rho g h \frac{\partial z_s}{\partial y} + \cos^2\theta_x \frac{g h^2}{2}\frac{\partial \rho}{\partial y} + \frac{\rho g n^2 U v}{\cos^{4/3}\theta_x h^{1/3}} = 0 \tag{3}$$

where t=time;x and y=longitudinal and lateral direction respectively; h=flow depth; u and v = flow velocity in x and y directions; $U = \sqrt{u^2 + v^2}$=resultant velocity; z_b and z_s=bed surface elevation and water surface above a reference datum respectively; n=Manning roughness coefficient; g=gravitational acceleration; ρ=density of the water and sediment mixture in the water column determined by $\rho = \rho_w(1 - C_t) + \rho_s C_t$, C_t=volumetric concentration of total-load sediment, ρ_s and ρ_w=density of sediment and water respectively; ρ_b=density of the water and sediment mixture in bed surface-layer determined by $\rho = \rho_w p'_m + \rho_s(1 - p'_m)$ p'_m=porosity of the surface-layer bed

material; θ_x is the angle of bed slope at longitudinal direct(x direct). Simulation of cohesive sediment transport process induced by embankment-break/breaching relates to cohesive material's erosion rate. Linear erosion function (Partheniades 1965) is used in this paper.

$$E_b = M\left(\frac{\tau_b - \tau_c}{\tau_c}\right) = k_d(\tau_b - \tau_c) \tag{4}$$

$$\tau_b = \frac{\rho n^2 g U^2}{h^{1/3}\cos\theta_x^{7/3}} \tag{5}$$

E_b=bed surface erosion rate; M=erodibility coefficient; k_d=detachment coefficient=M/τ_c; τ_b and τ_c =bed surface shear stress, critical shear stress respectively. Combine with Eq.(4) and Eq.(5) get sediment transport and bed change equations as below

$$\frac{\partial hC_t}{\partial t} + \frac{\partial huC_t}{\partial x} + \frac{\partial hvC_t}{\partial y} = k_d(\frac{\rho n^2 g U^2}{h^{1/3}\cos\theta_x^{7/3}} - \tau_c) \tag{6}$$

$$(1 - p'_m)\frac{\partial z_b}{\partial t} = -k_d(\frac{\rho n^2 g U^2}{h^{1/3}\cos\theta_x^{7/3}} - \tau_c) \tag{7}$$

the complete depth-averaged momentum equations and suspended-load transport equations contains turbulent stress and sediment flux term, as well as the non-uniform flow velocity induced dispersion term and sediment concentration profile over the depth (Wu 2007)

Avalanche Mechanism

In normal noncohesive material embankment, with the breach process continues the bed slope's angle gets larger. When the bed's angle exceed the repose angle, the noncohesive embankment soil slide in particles to the slope toe and form a new slope with the repose angle. However, in cohesive material(clay and slit) made embankment, the cohesive force between particles resists the sliding process. Instead of sliding process, mass failure due to safety factor of mass (Osman and Thorne 1988) happens.

$$R_s = \frac{f_r}{f_n}, R_s = \begin{cases} \text{stable} & \text{if } R_s > R_c \\ \text{unstable} & \text{if } R_s \leq R_c \end{cases} \tag{8}$$

f_r and f_d =resisting force and driving force respectively; R_s = the safety factor of the most dangerous bank mass, ratio of resisting force and driving force; R_c = critical safety factor, when R_s less than critical safety factor the bank is unstable then avalanche happens, vice versa; resisting force includes cohesion and frictional force while driving force is consisted of component force of mass weight as following:

$$f_d = W\sin\theta_y \tag{9}$$

$$f_r = W\cos\theta_y\mu + c'l \tag{10}$$

θ_y =bank slope's angle (y direction) μ=frictional coefficient; c' =cohesive coefficient; l=the length avalanche sliding plane ; W=the weight of the unstable bank mass, according to the relation of bed surface's density gets as below:

$$W = \rho_b g V = gV[p'_s \rho_s + (1 - p'_s)\rho_w] \tag{11}$$

Cohesive material embankment not only form steep slope bank, but also erode under the bank mass which is called the under cutting erosion. Here use the ratio of horizontal and vertical bed erosion to simulate lateral erosion as below:

$$E_h = R_e E_v \tag{12}$$

E_h=lateral(horizontal) bank erosion; E_v=vertical bed erosion; R_e=ratio of lateral erosion and vertical erosion. The corresponding cross section figure relates to avalanche process of this model are shows below:

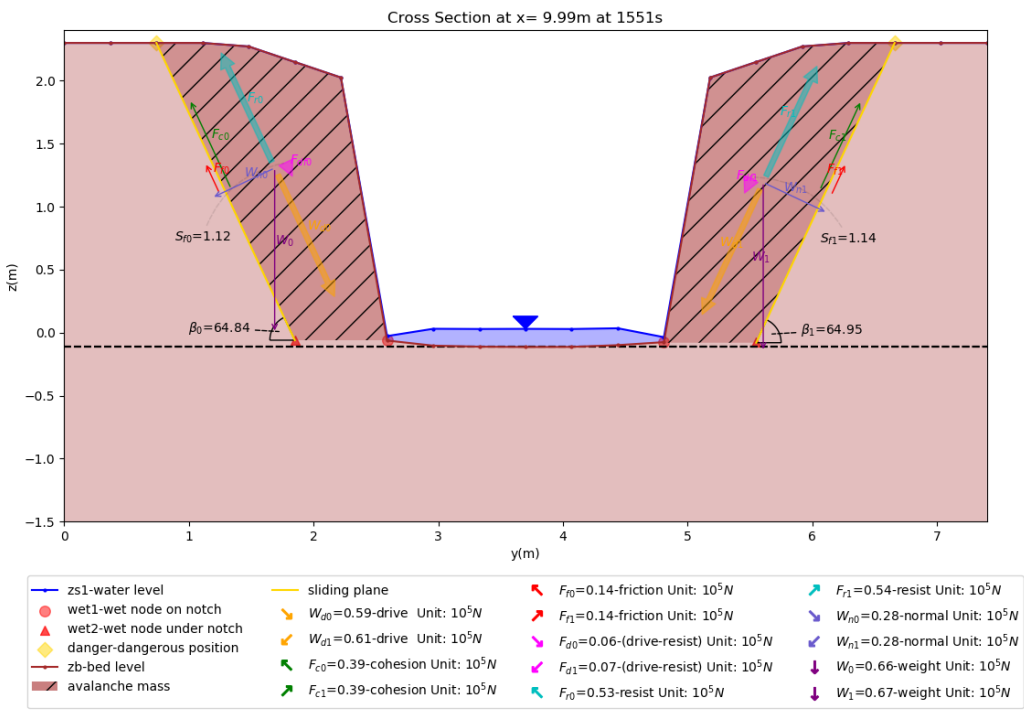


FIG. 1: Avalanche Analysis

The red triangle markers represent under cutting node, while round marker represent normal wet node. The brown mass is body of embankment, while the darker brown is the unstable bank mass. Yellow arrow and blue arrow are mass' driving force and resisting force respectively.

NUMERICAL METHOD

Finite-Volume Discretization

The Riemann solver based on the finite-volume method one of the popular methods to solve the embankment break and breaching flow problem that involves mixed flow regimes and hydraulic jump. Since the existing approximate Riemann solvers are usually developed for clear-water flow, Eq.(1) Eq.(2) Eq.(3) Eq.(6) , Eq.(7) are reformulated by moving the flow density from left sides to right sides by using the relations of density in water column and($\rho = \rho_w(1 - C_t) + \rho_s C_t$) bed surface($\rho = \rho_w p'_m + \rho_s(1 - p'_m)$) respectively. The derived continuity and momentum equations are

$$\frac{\partial h}{\partial t} + \frac{\partial uh}{\partial x} + \frac{\partial vh}{\partial y} = \frac{-E_b}{1 - p'_m} \tag{13}$$

$$\begin{aligned}\frac{\partial (hu)}{\partial t} + \frac{\partial (hu^2)}{\partial x} + \frac{\partial (huv)}{\partial y} = \sin\theta_x \cos\theta_x gh - \cos^4\theta_x \frac{gh^2}{2\rho}\frac{\partial \rho}{\partial x} - \cos^4\theta_x gh \frac{\partial z_s}{\partial x} \\ - \frac{gn^2 Uu}{\cos^{7/3}\theta_x h^{1/3}} + \frac{uE(\rho_s - \rho_w)}{\rho\cos\theta_x}\left(1 - \frac{c}{1 - p'_m}\right)\end{aligned} \tag{14}$$

$$\frac{\partial (hv)}{\partial t} + \frac{\partial (huv)}{\partial x} + \frac{\partial (hv^2)}{\partial y} = -\cos^2\theta_x \frac{gh^2}{2\rho}\frac{\partial \rho}{\partial y} - \cos^2\theta_x gh \frac{\partial z_s}{\partial y} - \frac{gn^2 Uv}{\cos^{4/3}\theta_x h^{1/3}} \tag{15}$$

The last terms on the right of Eq.(13) Eq.(14) Eq.(15) are derived from the flow density and bed change terms in Eq.(1) Eq.(2) Eq.(3) . vector form of Eq.(13) Eq.(14) Eq.(15) Eq.(6) are written as below:

$$\frac{\partial \Phi}{\partial t} + \frac{\partial F(\Phi)}{\partial x} + \frac{\partial G(\Phi)}{\partial y} = S(\Phi) \tag{16}$$

$$\Phi = \begin{bmatrix} h \\ hu \\ hv \\ hC_t \end{bmatrix}, \quad F(\Phi) = \begin{bmatrix} hu \\ hu^2 \\ huv \\ huC_t \end{bmatrix}, \quad G(\Phi) = \begin{bmatrix} hv \\ huv \\ hv^2 \\ hvC_t \end{bmatrix}$$

where Φ represent the vectors of unknown variables $F(\Phi)$, $G(\Phi)$ represent fluxes at x and y direction respectively and $S(\Phi)$ includes the remaining terms in each equation

Each control volume(cell) of finite-volume mesh is braced by four faces.A non-staggered (collocated) grid system is used here. The primary variables h, m, u, m, v and C_t are defined at the cell centers and represent the average values over each cell, while the fluxes are calculated at cell faces.

Integrating Eq.(16) over the (i,j) control volume, applying the Green theorem and using the Euler scheme for the time derivative, the following discretized equations are derived:

$$\Phi^{n+1}_{i,j} = \Phi^n_{i,j} - \frac{\Delta t}{\Delta x_{i,j}}(F^n_{i+1/2,j} - F^n_{i-1/2,j}) - \frac{\Delta t}{\Delta y_{i,j}}(G^n_{i,j+1/2} - G^n_{i,j-1/2}) + \Delta t S_{i,j} \tag{17}$$

where Δt=time step; $\Delta x_{i,j}$=x space step; $\Delta y_{i,j}$=y space step; $F^n_{i+1/2,j}$=flux at face(i+1/2,j); $G^n_{i,j+1/2}$=flux at face(i,j+1/2);

HLLC Riemann Solver and MUSCL Reconstruction

Solving the exact Riemann problem is rather complicated. The frequently used methods are approximate Riemann solvers such as HLL(Harten 1983) and HLLC(Toro et al. 1994). The model in this paper uses the HLL approximate Riemann solver, which is used to determine the fluid mass and momentum fluxes at cell faces.

$$F_{i+1/2} = \begin{cases} F_L & S_L \geq 0 \\ F_{hll} = \frac{S_R F_L - S_L F_R + S_R S_L (\Phi_R - \Phi_L)}{S_R - S_L} & S_L < 0 < S_R \\ F_R & S_R \leq 0 \end{cases} \tag{18}$$

where subscripts L and R denote the quantities evaluated at the left and right side of each cell interface, respectively. S_L and S_R represent the waves speeds which separate constant states of the local Riemann problem solution at cell interfaces. There are several ways to determine S_L and S_R, one choice was suggested by (Toro 2001) as follow:

$$S_L = u_L - a_L \lambda_L, S_R = u_R + a_R \lambda_R \tag{19}$$

where $u_K(K = L, R)$ = velocity at $K(= L, R)$ state; a_K =$\sqrt{gh_k}$ = wave celerity; and $lambda_K$=non unit coefficient is decided by

$$\lambda_k = \begin{cases} \sqrt{\frac{(h_* + h_k)h_*}{2h_k^2}} & \text{if } h_* > h_k \\ 1 & \text{if } h_* \leq h_k \end{cases} \tag{20}$$

h_*= estimate for the exact solution of h in the star region (Toro 2001), and can be evaluated as

$$h_* = \frac{1}{2}(h_l + h_R) - \frac{1}{4}(u_R - u_L)(h_L + h_R)/(a_L + a_R) \tag{21}$$

according to the Eq.(19) the S_L and S_R are calculated, then the left and right fluxes Eq.(18) are determined, so that the scheme has first-order accuracy in space. A second-order accurate approximation can be obtained by reconstructing the left and right states of fluxes using the monotonic upstream scheme for conservation laws(MUSCL) (Van Leer 1979). A piecewise linear reconstruction is used with the following slope limit method for gradient of state variable of flux f

$$\left(\frac{\partial f}{\partial x}\right)_i = \begin{cases} \max \begin{cases} 0 \\ \min(\beta\Delta_{i-1/2}, \Delta_{i+1/2}) \\ \min(\Delta_{i-1/2}, \beta\Delta_{i+1/2}) \end{cases} & \text{if } \Delta_{i+1/2} > 0 \\ \min \begin{cases} 0 \\ \max(\beta\Delta_{i-1/2}, \Delta_{i+1/2}) \\ \max(\Delta_{i-1/2}, \beta\Delta_{i+1/2}) \end{cases} & \text{if } \Delta_{i+1/2} < 0 \end{cases} \tag{22}$$

$$\Delta_{i-1/2} = \frac{f_i - f_{i-1}}{x_i - x_{i-1}}, \Delta_{i+1/2} = \frac{f_{i+1} - f_i}{x_{i+1} - x_i}$$

where the paramter β=1 reproduce the MINBEE (MINMOD) flux limiter, and the β reproduces the SUPERBEE flux limiter (Toro 2001).

Treatment of Bed Friction Term

The bed friction terms on the right sides of Eq.(14) and Eq.(15) are approximated by the following semi-implicit scheme:

$$gn^2 \frac{Uu}{\cos^{7/3}\theta_x h^{1/3}} = gn^2 \left(\frac{U}{\cos^{7/3}\theta_x h^{1/3}} \right)_{i,j}^{n} u_{i,j}^{n+1} \tag{23}$$

$$gn^2 \frac{Uv}{\cos^{4/3}\theta_x h^{1/3}} = gn^2 \left(\frac{U}{\cos^{4/3}\theta_x h^{1/3}} \right)_{i,j}^{n} v_{i,j}^{n+1} \tag{24}$$

and then move the $n+1$ time step value to the left side of Eq.(17) . This semi-implicit scheme is more stable than the fully explicit scheme for the bed friction terms. The θ_x represents the bed slope at x direction. Because the embankment is constructed with cohesive materials, it forms steep bed slope in flow stream wise direction. the bed slope in x direct accounts for the main effect of steep bed slope, therefor only contain θ_x in friction term. Since u velocity in x direction is affected by bed slope than y direction velocity v, it divided by $cos^{7/3}\theta_x$.

Treatment of Wet and Dry Node

During the flood propagation in floodplains nad channels with sloping boundaries, the water edges change with the time and computational nodes may become wet or dry. In the present model, a threshold flow depth(a small value such as 0.002m) is sued to decide drying and wetting. If the flow depth on a node is larger than the threshold value, this node is considered to be wet;otherwise, it is dry and flow velocity is zero.

Sediment Transport Computation

Eq.(17) is implemented to sediment transport Eq.(6) , the convective flux of sediment at cell faces is determined using hybrid linear/parabolic approximation (HLPA) scheme (Zhu 1991), which has approximately second-order accuracy in space.The quantities in the exchange term are evaluated using the values at cell centers at time level n. The bed deformation Eq.(7) is discretized as

$$\Delta z_b = \frac{k_d \Delta t}{1 - p'_m} \left(\frac{\rho n^2 g U^2}{h^{1/3} \cos\theta_x^{7/3}} - \tau_c \right)^n \tag{25}$$

The HLL approximate Riemann solver may provide a discontinuous solution of flow discharge near hydraulic jumps (Ying and Wang 2008). This may cause instability of sediment transport calculation because the bed surface erosion rate uses the bed shear stress and flow velocity that are obtained using the flow discharge. to avoid this, the flow velocity is determined from the inter cell fluxes as below:

$$\bar{u}_{i,j}^n = \frac{F_{h,i+1/2,j}^n + F_{h,i-1/2,j}^n}{2h_{i,j}^n}, \bar{v}_{i,j}^n = \frac{G_{h,i,j+1/2}^n + G_{h,i,j-1/2}^n}{2h_{i,j}^n} \tag{26}$$

where $\bar{u}_{i,j}^n$,$\bar{v}_{i,j}^n$= x velocity and y velocity from inter cell flux respectively; $F_{h,i+1/2,j}^n$=fluxes uh at cell faces between cell (i+1,j),(i,j); $F_{h,i-1/2,j}^n$=fluxes uh at cell faces between cell (i,j),(i-1,j); $G_{h,i,j/2+1}^n$=fluxes vh at cell faces between cell (i,j+1),(i,j); $G_{h,i,j/2-1}^n$=fluxes

vh at cell faces between cell (i,j),(i,j-1); the bed shear stress is then calculated from the above flow velocities using the Manning equation. The Manning's n for determing the bed shear stress is the same as that used in momentum Eq.(2) Eq.(3) Due to the Manning's n related the to hte bed form development that is not well understood in the case of embankment break/breaching flow, it is treated as a calibrated parameter in this study.

Implement of Avalanche

$$\Delta z^n_{h,i,j'} = R_e \Delta z^n_{b,i,j'} \tag{27}$$

where $x = i$ means at x node =i cross section, $\Delta z^n_{h,i,j_0}$,$\Delta z^n_{h,i,j_1}$ = n time step's horizontal side bank eroded length at (i, j_0),(i, j_1) node respectively; j_0,j_1 denote the side bank node at lower and higher y direction respectively; $\Delta z^n_{b,i,j_0}$,$\Delta z^n_{b,i,j_1}$ = n time step's vertical bed change at (i, j_0),(i, j_1) node respectively; R_e = ratio of horizontal erosion and vertical erosion of cohesive sediment;

where j' is the nodes at two side(left and right) of flow. Since this model is based on shallow water equation, it is a two-dimension model. However under cutting flow phenomenon in cohesive sediment is three dimensional problem. Here use virtual node and virtual side eroded length to solve this problem. Method are explained below:

$$\Delta Z_i^{\Delta t_0} = \sum_{t=s_0}^{t=e_0} \Delta z^t_{h,i,j_0}, \Delta Z_i^{\Delta t_1} = \sum_{t=s_1}^{t=e_1} \Delta z^t_{h,i,j_1} \tag{28}$$

$\Delta Z_i^{\Delta t_0}$ is the accumulated virtual erosion length during Δt_0 at node (i, j_0), and $\Delta t_0 = e_0 - s_0$. s_0, e_0 = start time and end time of lateral erosion respectively at lower y direction; On the other hand, $\Delta Z_i^{\Delta t_1}$ is the accumulated virtual erosion length during Δt_1 at node (i, j_1), and $\Delta t_1 = e_1 - s_1$, $\Delta t_1 = e_1 - s_1$ start time and end time of lateral erosion respectively at same x direction but higher y direction; Due to the calculation of accumulated virtual erosion length, the virtual nodes movement are determined as below:

$$(i, j_{0_v}) = (i, j_0 - \frac{\Delta Z_i^{\Delta t_0}}{\Delta y}), (i, j_{1_v}) = (i, j_1 + \frac{\Delta Z_i^{\Delta t_1}}{\Delta y}) \tag{29}$$

Where (i, j_{v_0}), j_{v_0}, (i, j_{v_1}), j_{v_1} are virtual nodes at lower and higher y direction respectively at same x$(= i)$ node. Δy= mesh space in y direction. At the initial embankment breach phases, virtual nodes are at same location as boundary wet nodes. With embankment break process continues, the side erosion (under cutting) gets severe which means virtual side eroded length gets longers and virtual nodes move deeper. This process continues until side mass avalanche happens. Then virtual eroded node overlap with normal wet node again, the virtual eroded length reset to zero as before.

model stability

Due to the developed solution procedure is explicit, the computational time step should be limited by numerical stability conditions, such as the Courant-Friedrichs-Lewy(CFL). Besides the numerical stability condition, additional conditions are needed

for sediment transport and bed change computation. This paper use the the bed change at each time step less than 10% of the the local flow depth h as an additional condition to get stable solution. In consequence, the time steps should satisfy the following condition:

$$\Delta t < min \begin{cases} \frac{\Delta x}{(|u|+\sqrt{gh})/\Delta x+(|v|+\sqrt{gh})/\Delta y} \\ \frac{0.1h(1-p'_m)}{k_d} / \left(\frac{\rho n^2 g U^2}{h^{1/3} \cos\theta_x^{7/3}} - \tau_c \right) \end{cases} \tag{30}$$

At the second part of inequality Eq.(30) is derived by limiting the bed change less than $0.1h$ in Eq.(27) . The $0.1h$ is an empirical estimated value through model testing and may vary in different cases.

MODEL TEST

The enhanced model was tested using USDA-ARS' large-scale overtopping failure tests on cohesive embankments (Hanson et al. 2005). The USDA-ARS Hydraulic Unit constructed a large-scale embankment testing facility in Saltwater, Oklahoma as following figure: The embankment 1 were seeded with fescue to protect against rainfall

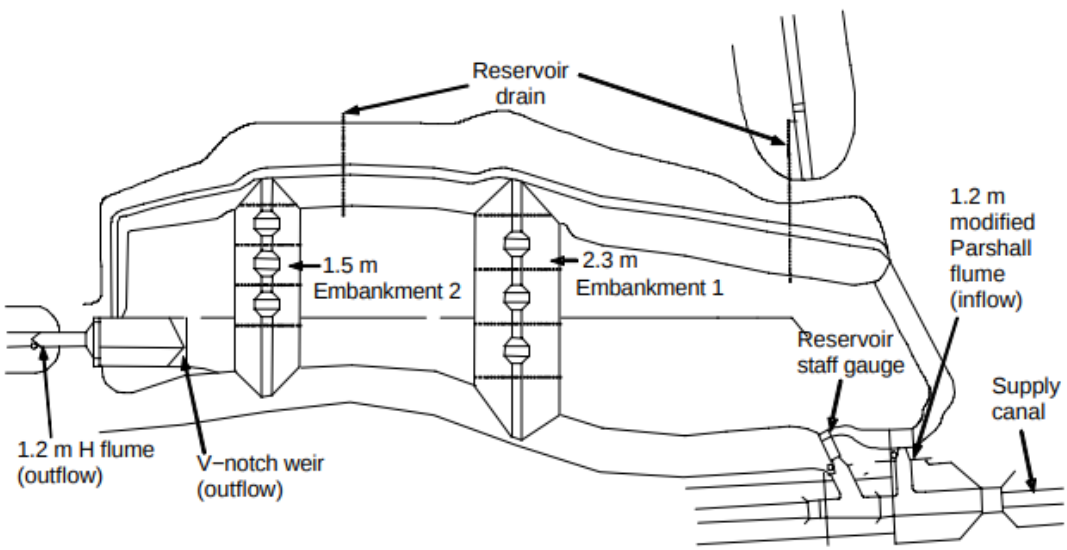
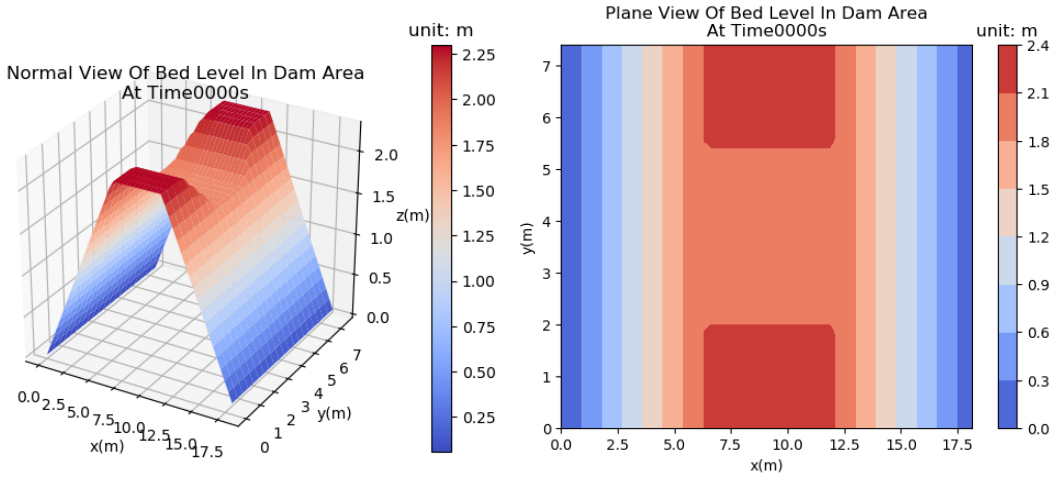


FIG. 2: Diagram of USDA-ARS Hydraulic Unit's Testing Facility

erosion prior to testing. Chart recorders were utilized to record inflow and outflow hydrography. An adjustable length(7 to 12 m) overhead rolling carriage with attached point gauge was utilized to obtain bed profiles, cross sections, and water surface elevations during testing Fig.2. Three large-scale cohesive embankment models, including two 2.3 m high and one 1.5 m high, were constructed and tested. The test case Embankment 1 was constructed 2.3 m height, having three test sections 7.3 m wide with different soil in each test section. Trapezoidal flow conveyance notches were cut into each test section to contain flow for the overtopping event. The dimensions of the

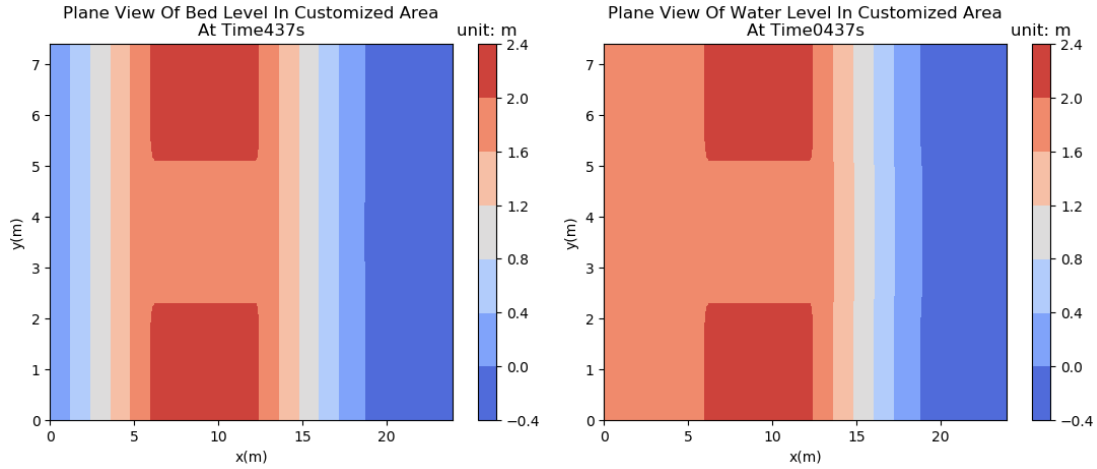
notches were 0.46 m deep by 1.83 m wide at the base for embankment 1. The embankment upstream and downstream faces, as well as notch side slopes, were 3 horizontal to 1 vertical. The sketch of Embankment shows as figFig.3.



(a) 3D view of Embankment 1

(b) Plane view of Embankment 1

FIG. 3: Sketch of Embankment 1



(a) Bed elevation over Embankment and its downstream at t=437

(b) Water surface elevation over Embankment and downstream at t=437

FIG. 4: Bed and water surface elevation over computational area at t=437

According to Embankment 1 constructed material type and related ratio (=70% sand, 25% silt and 5% clay), the Manning's n is between 0.017 to 0.025 (Chow 1959). In this test case, Manning's n = 0.018 is used. The tested embankment material is made

mainly by sand and silt, whose porosity p' is between 0.25 and 0.49 (Hough 1969). Based on the test dam's material belongs to sand and silt classification, corresponding cohesion is between 14 and 20 kpa (NAVFAC). And friction angle ϕ is between 27 to 33, as a result frictional coefficient μ is between 0.5 to 0.65 (Carter et al. 1991). Then present model chooses porosity $p' = 0.4$, cohesion $c' = 15kpa$, $\mu = 0.5$, summary property of model and soil are listed in Table.2:

Embankment	Value	Notch	Value
Height(m)	2.3	Depth(m)	0.46
Slope(h/v)	3:1	Slope(h/v)	3:1
Width(m)	7.4	Width(m)	1.83
Length(m)	18.13	Crest Length(m)	4.6

TABLE 1: Embankment and Notch Property

The computational domain are represent by a mesh consisting of 84×24 nodes in longitudinal and lateral directions. The longitudinal grid spacing was uniformly set as 0.365m in Embankment 1 region. The detachment coefficient k_d and cohesive sediment critical shear stress τ_c are set as $10.3 \times 10^{-5} m^2 s/kg$ and $0.14Pa$ respectively.Both values are obtained from the soil property's section of USDA -ARS' test report (Hanson et al. 2005). The time step length varies with time as determined using inequality Eq.(30) .

Model	Value	Soil	Value
Mesh	84×21	Erode Coefficient($m^2 s/kg$)	10.3×10^{-5}
Domain(m^2)	410×7.4	Critical Shear Stress(pa)	0.14
Embarkment area(m^2)	24.89×7.4	Porosity	0.4
x direct's grid space(m)	0.365	Cohesion(pa)	15000
y direct's grid space(m)	0.365	Frictional Coefficient	0.5

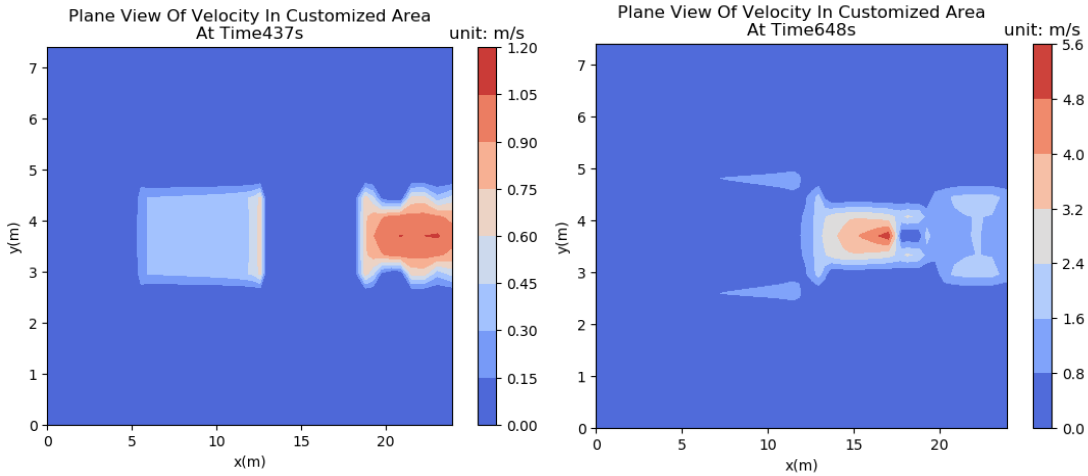
TABLE 2: Model and Soil Property

Embankment breaching Stage

Initial Stage

The embankment 1 were seeded with fescue to protect against rainfall erosion prior to testing. Inflow to the reservoir during the testing was supplied by canal and measured at test site with a modified Parshall flume for embankment 1. The reservoir upstream of the test embankments was filled in advance of test to a depth of 1.2m. Therefore the initial stage of overtopping process starts at $t_0 = 437s$.

Fig.5a is the simulated depth-averaged velocity distribution at $t = 437$, which shows flow's velocity at dam's notch and downstream dam toe(x=20m). Similarly, Fig.5b is the velocity figure at $t = 648s$, illustrates that flow's velocity is faster at two side of the notch and dam toe, but is zero at the center of downstream. This is because

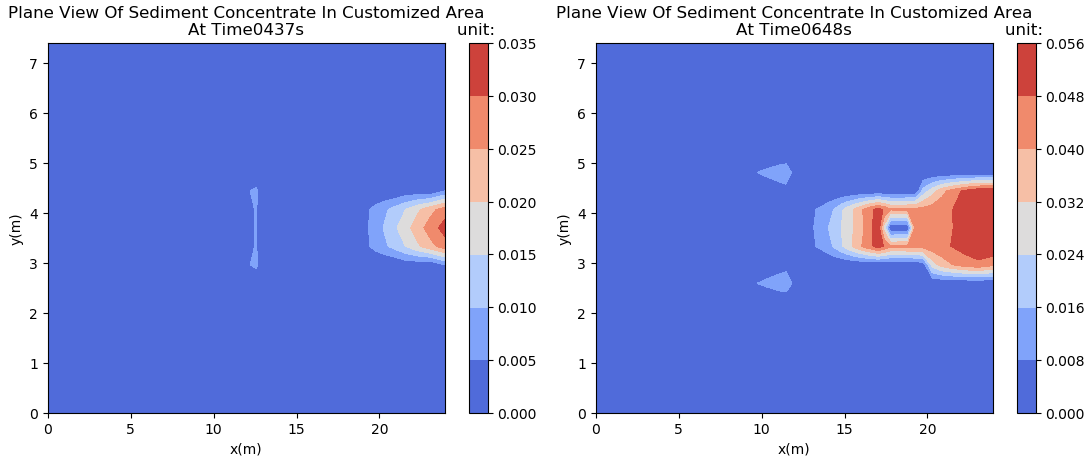


(a) Velocity over Embankment and its downstream at t=437s

(b) Velocity over Embankment and its downstream at t=648s

FIG. 5: Velocity at initial stage of the Embankment 1 breaching

Initial overtopping flow results in sheet and rill erosion with one or more master rills developing into a series of cascading overfalls. This model shows 2 rill erosion along the two sides of notch.



(a) Sediment Concentration over Embankment and its downstream at t=437s

(b) Sediment Concentration over Embankment and its downstream at t=648s

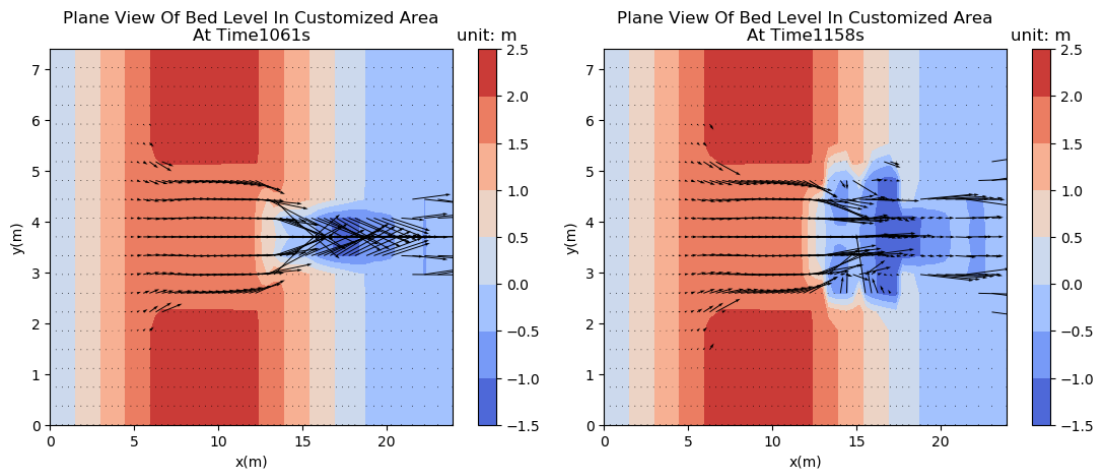
FIG. 6: sediment concentration at initial stage of the Embankment 1 breaching

Fig.6a is the sediment volumetric concentration figure at $t = 437$, reveal that the higher concentration at embankment downstream which is related to flow velocity. On the other hand, Fig.6b shows the sediment concentration distribution at $t = 648$.It

demonstrates that the sediment concentration is higher at dam toe and downstream lower at the center of dam downstream toe, which is similar to velocity distribution.

Avalanche Stage

As erosion continues, previous rill erosion at the two side of bank (Fig.5a) forms the master rill (open channel) and develop into the overfall. In addition to embankment bed's vertical bed erosion, flow also erodes the two side bank of the formed channel's laterally. Under the the effect of vertical and lateral erosion (under cutting), both sides of the eroded channel was shaped to steep sided gorge from. According to Fig.1 and Eq.(8) , When the safety factor R_s of the overhang mass at its channel side less than critical safety factor R_c, avalanche happens.



(a) Bed elevation over Embankment and its downstream at t=1061s

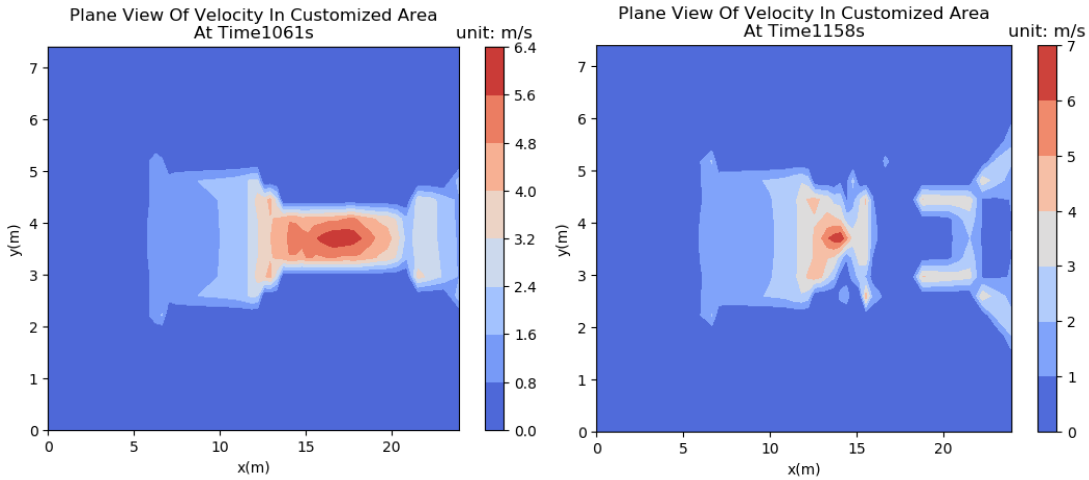
(b) Bed elevation over Embankment and its downstream at t=1158s

FIG. 7: Bed elevation over computational area at avalanche stage

Fig.7a is embankment downstream toe's erosion condition at t=1061s. The vertical and lateral erosion process continues, resulting safety factors of two side banks become smaller. The most dangerous mass at the bank are still stable, until t=1158s Fig.**??** the risky mass collapse at embankment downstream toe (x=17.5m).

Fig.8a illustrates the velocity distribution before avalanche (at t=1061s,) which shows the fluent flows fastest at the embankment downstream toe. However, after avalanche (at t=1159s) the velocity decreases most at downstream toe based on Fig.8b. This is because the channel bed enlarged duet to mass failure at the downstream toe. This phenomena causes the flow turn into subcritical flow from critical flow at the mass collapsed region, leading to decreasing velocity. Moreover, Fig.8b shows the flow moves fast before avalanche position (x= about 14m), and it indicates the erosion continues.

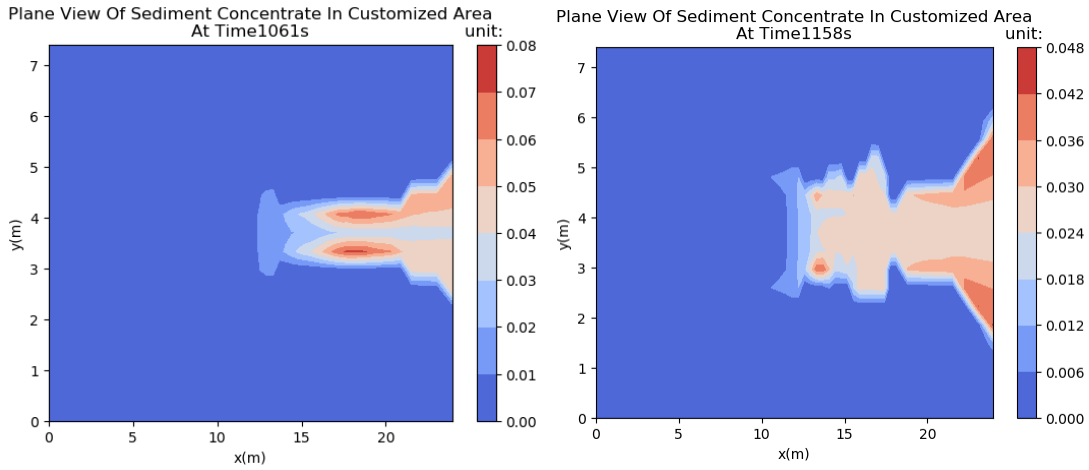
After analysis of embankment bed elevation and velocity distribution, Fig.9a, Fig.9b shows the Sediment volumetric concentration condition before the avalanche and after avalanche respectively. Fig.9a shows that sediment concentration is highest at the



(a) A velocity over Embankment and its downstream at t=1061s

(b) A velocity over Embankment and its downstream at t=1158s

FIG. 8: velocity at downstream eroded stage



(a) Sediment Concentration over Embankment and its downstream at t=1061s

(b) Sediment Concentration over Embankment and its downstream at t=1158s

FIG. 9: velocity at downstream eroded stage

both sides of downstream slope channel. Nevertheless, the velocity, affects the erosion process and sediment concentration most, reaches the fastest at center of downstream slope channelFig.8a. The reason of situation of difference is channel bank's lateral erosion's contribution to sediment transport is larger than the peal velocity at the center of channel. Fig.9b shows that overall sediment concentration is lower than before, while the collapsed area's sediment concentration gets higher.

Mass Failure Analysis

Instead sediment particle slides with its repose angle in non-cohesive material embankment, cohesive embankment's breaching mechanism is bank mass failure. Eq.(8) informs that the bank mass collapse,the bank mass collapse, when its safety factor is less than critical safety factor.

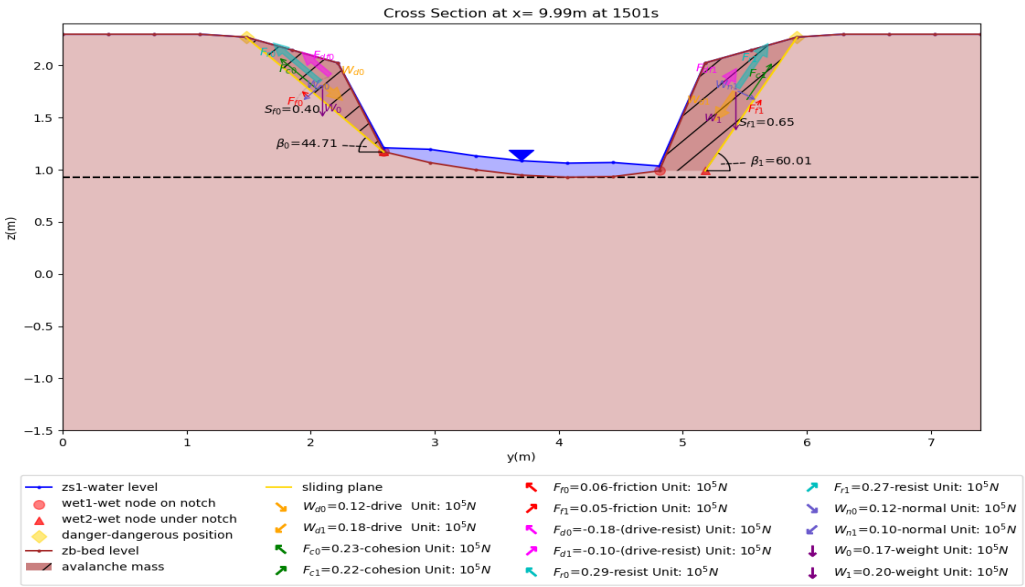


FIG. 10: Bed elevation over Embankment and its downstream at t=1501

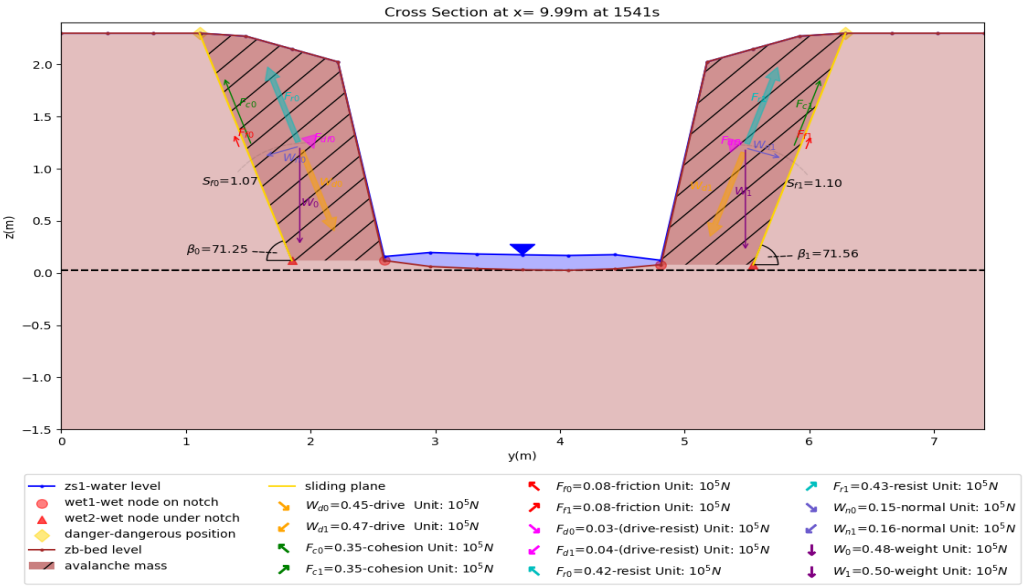


FIG. 11: Bed elevation over Embankment and its downstream at t=1541

Fig.10 is the breaching formation figure including bank mass's resist and driving force at x=9.9m at t=1501s. This shows left bank's virtual node is at the same node as wet node, while right bank's virtual node cuts to the right of the wet node. With the wet nodes and virtual nodes, it forms two potential sliding plane(yellow line), and the corresponding most dangerous bank mass(dark brown). The left potential sliding plane's angle $\beta_0 = 44.7$, driving force $W_{d0} = 12000N$, resultant force $F_{d0} = -18000N$ (driving force-resisting force), safety factor $R_{s0} = 1/0.4$. Correspondingly, right potential sliding angle $\beta_1 = 60$, driving force $W_{d1} = 18000N$,resultant force $F_{d1} = -10000N$ (driving force-resisting force), safety factor $R_{s1} = 1/0.65$. This is because, right bank's erosion is severer than left bank. However, bank's mass are still in stable condition.

Fig.11 is the breaching formation figure at same position at t=1541s. Both bank's virtual nodes cut deeper than wet nodes, which form the most dangerous mass angle $\beta_0 = 70.25$,$\beta_0 = 71.56$ respectively. and the resultant force of both sides are $F_{d0} = 300N$,$F_{d1} = 400N$ respectively. Equally, the left and right bank's safety factor are $R_{s0} = 1/1.07$, $R_{s1} = 1/1.1$ respectively. Even the safety factors are smaller than before but, it larger than R_c critical safety factor.

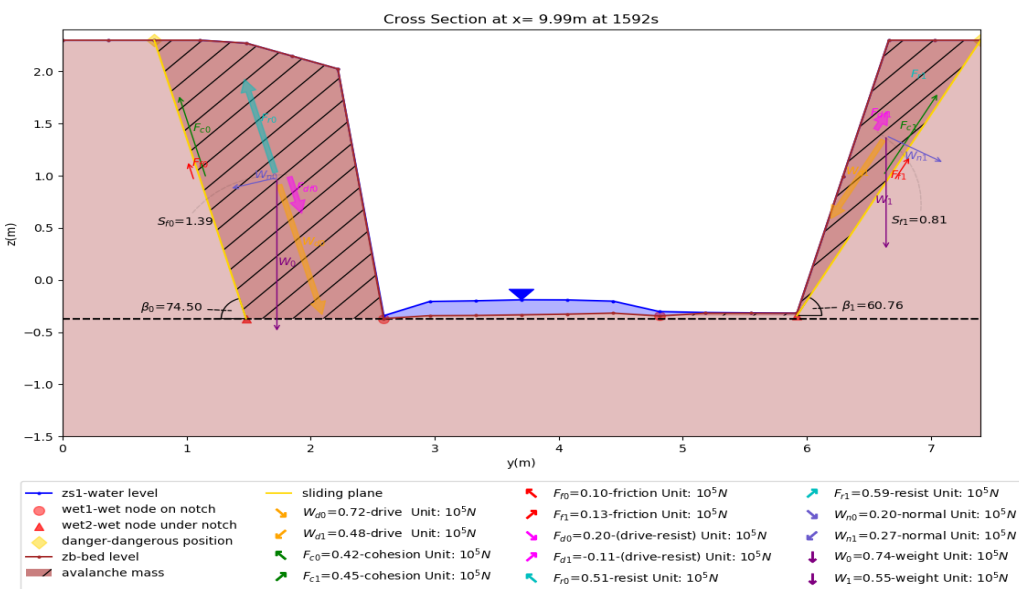


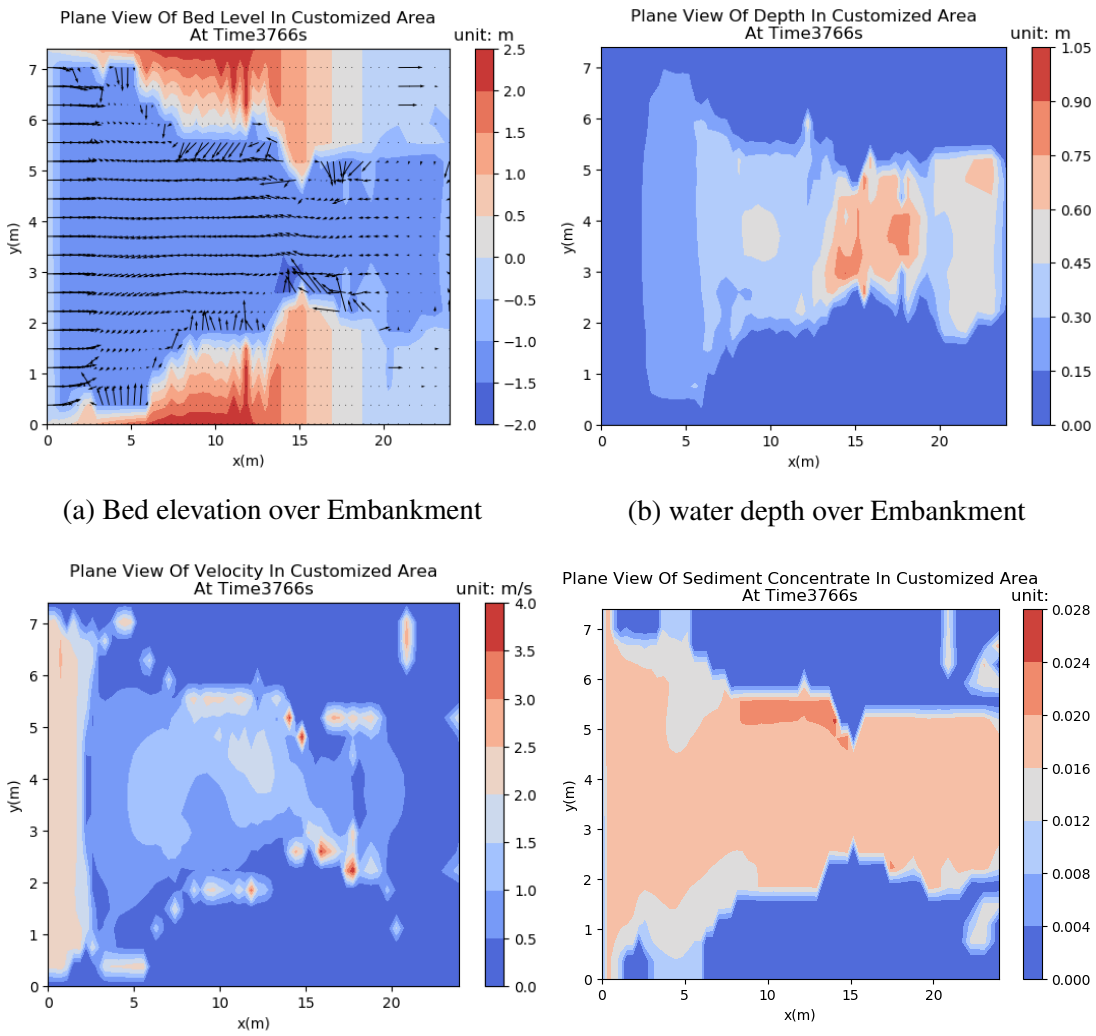
FIG. 12: Bed elevation over Embankment and its downstream at t=1592

Fig.12 is the after avalanche figure at t=1592s. According to the image, the left side bank is quit unstable. Actually, its safety factor has already reached the critical safety factor but it still overhangs there. The reason is that the mass avalanche algorithm is not for a single slice, it considers the mass body which is consisted of its containing slice together. Hence, even the slice's bank mass at x=9.99m is unstable, its neighboring slice ,which is at stable condition, restrain its collapse movement.

On the other hand, the avalanche happens on the right side bank with it neighboring

bank body. The collapsed mass was washed away by the flow, and remained the channel bed at x=9.99m as the same level as right wet node. The algorithm then focuses to calculate the further node of right bank's safety factor, and illustrate that side mass is stable now.

Final breachinsgs Stage



(a) Bed elevation over Embankment

(b) water depth over Embankment

(c) flow velocity over Embankment

(d) sediment concentration over Embankment

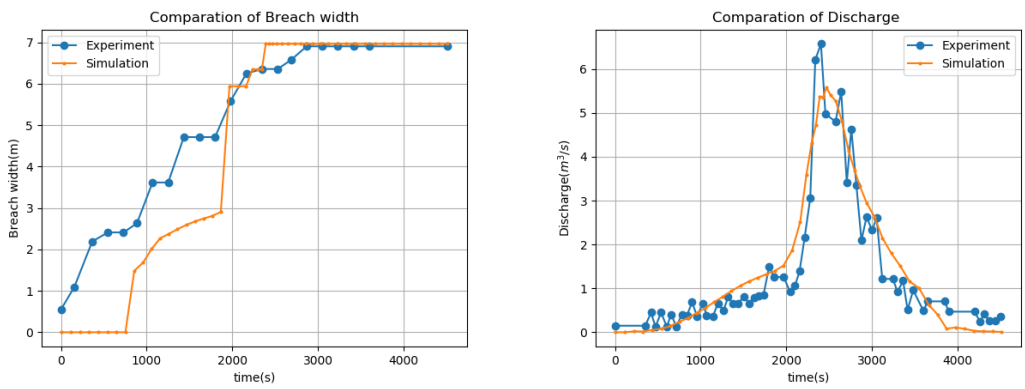
FIG. 13: Bed level, Depth, Velocity, sediment concentration at t=3766s

Embankment's breaching mechanism is alternant process of erode and bank mass collapse. Flow erodes bed and bank to form unstable side slope, then unstable mass collapse and side slope becomes table, then side slope erode again. According to (Hanson et al. 2005), at t=3060s the inflow was diminished to zero, then the cohesive embankment erode process ends.

Fig.13a is the bed elevation at final stage. This figure shows that the backflow happens at some low-land region. This is because the upstream inflow diminished and the water flows to lower elevation by gravity. According to the figure, there is a serve eroded low-land near embankment's downstream corner, where backflow happens most. Fig.13b shows depth(like h=1m) are higher between x=13m to x=17m than other region's depth(h¡0.6m). This is because, x=13m-17m area is dam downstream slope area, where erode most.

Fig.13c informs that velocity in stream wise at its main channel decreased to $2(m/s)$, while backflow's velocity at the banks side reaches $4(m/s)$ due to the lower-land. Fig.13d illustrates that sediment concentration distribute more uniformly than the previous breaching stage. The sediment concentration is little higher at higher side bank region(x=8m-15m, y=5m-6m). And overall sediment transport is restrained.

Comparison with Experiment Result



(a) Comparison of measured and calculated breach width

(b) Comparison of measured and calculated discharge

FIG. 14: Comparison with Experiment

After the simulation of cohesive embankment overtopping and breaching process, the recorded discharge and breaching width are compared with field test's result (Hanson et al. 2005), showing in Fig.14a and Fig.14b.

shows that during the initial stage(before t=800s) the measured breaching width is larger than simulation result. This is because the USDA observed erosion at that stage is in the form of sheet and rill, it is hard to simulate. Since the embankment soil composition and structure is uncertain, and rill and sheet erosion happens at random location. Therefore, before t=800s, the calculated width is less than measured.

From t=1000s to t=1900s the width breach width enlarges by vertical erosion's region gets wider. The calculated width increasing tendency is similar to measure result at this stage. During t=1900s to t=2000s the simulated breaching width increased from 3m to 6m, while the experiment's breaching width raise from 4.8m to 6.2m during t=1900 to t=2200. And the mechanism of enlarging width at both model simulation

and experiment is the mass avalanche. However, the increasing rate of width is different. This is because the model's avalanche algorithm estimates safety factor and make collapse immediately. In contrast, field test's avalanche process proceed slowly. Finally, calculated and measured breach width reaches the largest $w = 7m$ at t=2400s and t=2800s respectively.

Fig.14b demonstrates that during the initial stage(before t=800s) the measured discharge is ups and downs, the embankment downstream slope was eroded in sheet and rill form. This is because the water flows to downstream slope from notch first (discharge increase), then water flows into immersible vegetal layer (discharge increase). This process continues ,which accounts for the change of discharge at the initial stage, until form one or more master rills. After t=800s simulated discharge increases steadily, and match the measured results well. After t=2000s the calculated discharge increase sharply and reaches the peak discharge $q = 5.6m^3/s$ at t=2469s. On the other hand, the measured discharge increased even faster to peak discharge $q = 6.5m^3/s$ at t=2405s. Hence, the simulated peak discharge time is later 2% than the experiment, and the peak discharge is 13.8% less than measured results.

CONCLUSION

A depth-averaged 2D finite-volume model has been developed to simulate hte flow, sediment transport and morphological changes induced by embankment break/breaching. The model uses the improved shallow water equations that takes into account the effect of sediment concentration, bed change and steep bed slope on the flow. The sediment transport adopts total-load method and assume all sediment is from eroding the embankment body and wash away immediately. The governing equations are solved using an explicit finite-volume method with the HLL approximate Riemann solver to handle the mixed-regime flows of embankment break/breaching and wetting-drying problem effectively. This method is first-order accuracy in time and second-order accuracy in space. A specially designed mass collapse algorithm is used in this model, to consider that lateral erosion (undercutting phenomena)besides the vertical erosion. The model's time step is varied by satisfying the CFL stability condition and less than 10% of local flow depth.

The enhanced model was tested by using the USDA-ARS filed test in Stillwater, Oklahoma. After comparing with measured data, present model predicts generally well the breach width and flow discharge. The calibrated parameter such as threshold flow depth, Manning coefficient and critical safety factor have different value in case of dam/dike breaching and avalanche, perhaps due to different flow conditions and embankment configuration. The model's result indicates

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